# Cutover runbook: amsat.org/status → status.amsat.org

Merging PR #30 (Issue #24: report timestamp precision) as part of the move from dallas190 to azure193. Window: **Wednesday 4 Nov 2026, ~14:00 UTC** (21:00 Bangkok).

**AS RUN, 4 Nov 2026: completed.** status.amsat.org is live at API 1.2; the 308 redirect is on. Deviations from this plan: (1) the migration's first run failed on rows dated 1970-01-01 (below the TIMESTAMP minimum). The dump was reloaded, 14 such rows deleted, and the re-run succeeded; the script now has this as step 0. (2) Hotfix #51 (php.ini-production) after the first test report showed PHP warnings. (3) #30 also shipped corrections and a `correction` field (closing #31), so where this checklist says "replaced_count removed" or "no replace-on-repeat", see the CHANGELOG instead. Full record: `docs/runbooks/2026-11-04-cutover.md`.

## How the pieces fit

**dallas190 is never changed.** You take a copy (dump) of its database exactly as it is, with the old schema. You load that copy into azure193 and run #30's migration *there*. The new columns, the backfill and the new indexes exist only on azure193. dallas190 keeps the old schema and old code, so it stays a working fallback.

| | Where | What | Who / how |
|---|---|---|---|
| 1 | dallas190 | Freeze writes, take a dump | You, by hand |
| 2 | azure193 | Load the dump (old schema), run the migration | You, by hand |
| 3 | GitHub | Merge #30 | You; CD then deploys automatically |
| 4 | status.amsat.org | Smoke test the new code on the migrated data | You |
| 5 | Cloudflare | Enable the drafted 308 redirect rule | You |
| 6 | Everyone | Verify, announce, monitor | You |

### What merging #30 does by itself

• CD runs the full CI gate (PHPUnit, Playwright, compose smoke test, migration dry run), then builds and pushes both images and deploys to azure193.
• The deploy ships the new `deploy/docker-compose.yml` and recreates the containers. The DB container restarts once with `--default-time-zone=+00:00` (data volume kept), and the frontend gets `API_INTERNAL_URL`.
• New code reads and writes `observation_time` / `submission_time`; the form posts through the API.
• API version becomes **1.1**. The API breaking changes go live (removed `period`, real `reported_time`, and `replaced_count` replaced by a `correction` boolean; repeat reports are kept, with the grid and summary counting each callsign's latest).
• It does **not** run the migration. That's step 2, and it must happen first.

**Order matters:** migrate *before* merging. If #30 deploys onto the old schema, the new code fails on every page load and submission until the migration runs.

## A. Before the window (by 3 Nov)

- [ ] **#30 is green and mergeable.** Read the final diff once more. No other open PRs touch `CHANGELOG`, `db/` or `api/v1/`.
- [ ] **Merge #48** (docs-only usage-guideline clarifications from #47) before the window. It touches the same overview section and the CHANGELOG as #30, so #30 gets updated onto it afterwards. Confirm #30 is still green and mergeable after that.
- [x] **Decide #24 closure:** add "Closes #24" to #30's description, or close #24 by hand after cutover. **Done 1 Nov:** #30's description says "Closes #24", so merging #30 closes it automatically.
- [x] **Check dallas190's database engine and table definition. Done 1 Nov:** MariaDB 10.11.14 (Ubuntu 24.04), database `amsat_status`. The `satellite` table is identical to the snapshot the CI migration dry run uses, and day/hour/period are nullable. A MariaDB 10.11 dump loads into 12.3 as is.
- [x] **Identify the app user (needed for the write freeze in B1). Done 1 Nov:** the "app user" (the MariaDB account the status page's PHP code logs in as, used by the form, the admin pages and the API on dallas190) is `'amsatstatus'@'localhost'`. Its grants:

**REFERENCE · SHOW GRANTS OUTPUT (DON'T RUN)**

```
GRANT USAGE ON *.* TO `amsatstatus`@`localhost` ...        -- no server-wide rights
GRANT ALL PRIVILEGES ON `amsat_status`.* TO `amsatstatus`@`localhost`
```

This is the ideal shape: the B1 revoke touches only the status database, and the rollback is one `GRANT ALL`. Confirmed 1 Nov: the account exists only for `localhost` (no second host entry that would keep write rights).

- [x] **Confirm access:** SSH to both hosts, a DB account on dallas190 that can REVOKE/GRANT, a way to copy a file from dallas190 to azure193, Cloudflare dashboard, and GitHub merge rights. **Done 1 Nov.**
- [ ] **Rehearsal (recommended):** dump dallas190 now, load it into a *throwaway* MariaDB 12.3 container on azure193 (separate from production; azure193's real DB container has a random root password by design, so it can't host a scratch database), run the migration there, time it, and delete the container. Real data and real hardware: it checks the load and gives true timings for the window.

**SHELL · DALLAS190**

```
sudo mysqldump --single-transaction --no-tablespaces amsat_status \
  | gzip > ~/amsat_status-rehearsal.sql.gz
ls -lh ~/amsat_status-rehearsal.sql.gz
# then copy it to azure193, e.g. via your laptop:
#   scp dallas190:amsat_status-rehearsal.sql.gz . && scp amsat_status-rehearsal.sql.gz azure193:
```

**SHELL · AZURE193**

```
cd ~
RAW=https://raw.githubusercontent.com/AMSAT-NA/satellite-status\
/issue-24-observed-at-precision/db/migrations
curl -fsSLO "$RAW/2026_08_08_add_observation_time_submission_time.sql"
docker run -d --name rehearsal -e MARIADB_ROOT_PASSWORD=rehearse mariadb:12.3
until docker exec rehearsal mariadb -uroot -prehearse -e 'SELECT 1' >/dev/null 2>&1; do sleep 2; done
docker exec rehearsal mariadb -uroot -prehearse -e 'CREATE DATABASE r'
time (gunzip -c amsat_status-rehearsal.sql.gz | docker exec -i rehearsal mariadb -uroot -prehearse r)
# migrate (prints the pre-flight counts first)
time docker exec -i rehearsal mariadb -uroot -prehearse r < 2026_08_08_add_observation_time_submission_time.sql
docker exec rehearsal mariadb -uroot -prehearse r -e "
  SELECT COUNT(*) FROM satellite;
  SELECT COUNT(*) FROM satellite WHERE observation_time IS NULL;     -- 0
  SELECT COUNT(*) FROM satellite WHERE submission_time IS NOT NULL;  -- 0
  SHOW INDEX FROM satellite WHERE Key_name LIKE 'idx_satellite%';"
docker rm -f rehearsal
```

`sudo mysqldump` logs in as MariaDB root through Ubuntu's socket login, so there's no password prompt. `--single-transaction` takes a consistent copy without locking the live site.

Rehearsal: load time / migration time _____

- [x] **Cloudflare:** the 308 rule is drafted (disabled). Test its matching in the rule preview/trace tool:

  **CLOUDFLARE RULE EDITOR**

  ```
  (http.host in {"amsat.org" "www.amsat.org"} and (http.request.uri.path eq "/status"
    or starts_with(http.request.uri.path, "/status/")))
  → wildcard_replace(http.request.uri.path, "/status*", "https://status.amsat.org${1}")
     308, preserve query string ON
  ```

  **Done 1 Nov:** live-tested behind a `cutovertest=1` guard. Bare path and API path with query both 308 to the right URL, `/statusreport` and unguarded `/status/` were not redirected, and a POST survived the redirect (308 then 422 `missing_name` from status.amsat.org). Rule left disabled with the guard removed.

- [x] **Announce to API consumers** (Reiner/Ionaut and the other app developers): the date, the new base URL `https://status.amsat.org/api/v1/`, and the #30 breaking changes. No field renames: `reported_at` in, `reported_time` out. **Done 1 Nov:** site banners, email listserv, Discord.

- [x] **Site banner** on dallas190 shows the date/time and new URLs. Optionally add one line about the API breaking changes. **Done 1 Nov.**

- [ ] **Stage the window files** on the right hosts (drafted 1 Nov): `banner-maintenance-dallas190.md` and `submit-paused.php` in your home directory on dallas190, `banner-welcome-azure193.md` on azure193. Edit the expected finish time in the maintenance banner if your rehearsal timing suggests otherwise.

- [x] **Confirm the `legacy-bridge` tag exists** (code running on dallas190), for reference during rollback. **Done:** tag points at `c9b310a`.

## B. During the window: before merging #30

Window start (UTC) _____

### B1. Freeze dallas190

- [ ] Swap in the maintenance banner on dallas190:

  **SHELL · DALLAS190**

  ```
  cp banner-maintenance-dallas190.md /var/www/www.amsat.org/status/content/banner.md
  ```

- [ ] **Pause the form handler** (recommended). Once writes are revoked, the old `submit.php` on dallas190 (PHP 8.3.6) throws a database exception, so visitors get a blank page or a 500 error. The stub tells them plainly that their report wasn't saved and where to go:

  **SHELL · DALLAS190**

  ```
  cd /var/www/www.amsat.org/status
  sudo cp -p submit.php ~/submit.php.cutover-bak       # backup OUTSIDE the web root
  sudo cp ~/submit-paused.php submit.php               # overwrite with the paused version
  sudo chown --reference=index.php submit.php          # match owner/permissions
  sudo chmod --reference=index.php submit.php
  ```

  Keep the backup out of the web root: Apache would serve a `.cutover-bak` file as plain text. Drop sudo if you own the web files. No restore needed after a successful cutover, because the redirect sends everyone to the new site. Restore only on rollback (section G).

- [ ] **Stop writes** to the status database only, by taking write rights away from the app user `amsatstatus`. Run as an admin in MariaDB on dallas190. Nothing else on the server is affected.

  **MARIADB CLIENT · DALLAS190 (SUDO MYSQL)**

  ```
  REVOKE INSERT, UPDATE, DELETE ON amsat_status.* FROM 'amsatstatus'@'localhost';
  SHOW GRANTS FOR 'amsatstatus'@'localhost';
  -- ALL PRIVILEGES is now listed as individual rights, without INSERT, UPDATE, DELETE
  ```

  Takes effect for new connections. PHP opens a fresh connection on every page load, so in practice it's immediate. Reading still works, so the old site keeps displaying during the window. **Undo (rollback only):**

  **MARIADB CLIENT · DALLAS190 (SUDO MYSQL)**

  ```
  GRANT ALL PRIVILEGES ON amsat_status.* TO 'amsatstatus'@'localhost';
  ```

- [ ] Confirm the freeze: the form now shows the "Reports paused" page, and an API POST to dallas190 returns `500 write_failed`:

  **SHELL · ANYWHERE**

  ```
  curl -s -X POST https://www.amsat.org/status/api/v1/reports.php -H 'Content-Type: application/json' \
       -d '{"name":"AO-91","report":"Heard","callsign":"W1AW","reported_at":"2026-11-04T13:00:00Z"}'
  # {"error": {"code": "write_failed", ...}}   (nothing is stored)
  ```

### B2. Dump dallas190

- [ ] Record the row count, then dump. Don't use `--databases`, so the dump loads into whatever database name azure193 uses.

  **MARIADB CLIENT · DALLAS190 (SUDO MYSQL)**

  ```
  SELECT COUNT(*), MAX(id) FROM amsat_status.satellite;
  ```

  **SHELL · DALLAS190**

  ```
  sudo mysqldump --single-transaction --no-tablespaces amsat_status \
    > ~/satstatus-dallas190-$(date -u +%Y%m%dT%H%MZ).sql
  ls -lh ~/satstatus-dallas190-*.sql
  ```

  dallas190 row count / MAX(id) _____

- [ ] Keep the dump file. It's also your pre-migration backup. Copy it to azure193 (and somewhere safe off both hosts).

### B3. Load into azure193 (old schema)

- [ ] On azure193, define a one-line db helper for this shell session. It runs SQL in the production DB container using the container's own credentials, so no password is typed. Every db command below uses it.

  **SHELL · AZURE193**

  ```
  cd /opt/services/satellite-status
  db() { docker compose exec -T db sh -c \
    'mariadb -u"$MARIADB_USER" -p"$MARIADB_PASSWORD" "$MARIADB_DATABASE"'; }
  ```

- [ ] Load the dump into the production DB container. This replaces azure193's current test data. The old-code containers can stay up; there's no traffic yet.

  **SHELL · AZURE193**

  ```
  cd /opt/services/satellite-status
  db < satstatus-dallas190-<stamp>.sql
  ```

- [ ] Row count on azure193 matches dallas190:

  **SHELL · AZURE193**

  ```
  echo 'SELECT COUNT(*) FROM satellite;' | db
  ```

### B4. Migrate on azure193

- [ ] Fetch both migration files from #30's branch onto the host (they aren't there until #30 deploys):

**SHELL · AZURE193**

```
cd /opt/services/satellite-status
RAW=https://raw.githubusercontent.com/AMSAT-NA/satellite-status\
/issue-24-observed-at-precision/db/migrations
curl -fsSLO "$RAW/2026_08_08_add_observation_time_submission_time.sql"
curl -fsSLO "$RAW/2026_08_08_add_observation_time_submission_time_rollback.sql"
```

- [ ] **Pre-flight:** run the SELECT block near the top of the migration on its own and read the counts:

**SHELL · AZURE193**

```
echo "SELECT SUM(day IS NULL) null_day, SUM(hour IS NULL) null_hour,
  SUM(period IS NULL) null_period,
  SUM(period IS NOT NULL AND period NOT BETWEEN 0 AND 3) bad_period,
  COUNT(*) total_rows FROM satellite;" | db
```

Expect `null_day = 0`, `null_hour = 0`, `null_period + bad_period ≈ 53` (the known `period = -1` rows), and `total_rows` = the B2 count. Anything very different: stop and investigate before migrating.

- [ ] **Run the migration** (it pins UTC, adds columns, backfills, enforces NOT NULL, adds both indexes, then sets the `submission_time` default):

**SHELL · AZURE193**

```
time db < 2026_08_08_add_observation_time_submission_time.sql
```

Migration duration _____

### B5. Verify the migrated database

- [ ] All four checks pass:

**SHELL · AZURE193**

```
echo "SELECT COUNT(*) FROM satellite;                                  -- = B2 count
SELECT COUNT(*) FROM satellite WHERE observation_time IS NULL;    -- 0
SELECT COUNT(*) FROM satellite WHERE submission_time IS NOT NULL; -- 0 (historical rows)
SHOW INDEX FROM satellite;  -- lists both idx_satellite_* indexes
" | db
```

- [ ] Optional: drop the leftover `satellite_old` table that came over in the dump. It's no longer in `db/schema.sql`, and dallas190 still has it.

**SHELL · AZURE193**

```
echo 'DROP TABLE IF EXISTS satellite_old;' | db
```

## C. During the window: merge #30

- [ ] Merge PR #30 (squash) on GitHub.
- [ ] Watch the CD run in Actions: **CI Gate** (including *migration-dry-run*) → **Docker Build Test** → **Push to ACR** → **Configure Server Environment** → **Deploy**, all green.

#30 merge commit SHA _____

Deploy finished (UTC) _____

- [ ] Containers are up, and the DB container now runs with UTC:

**SHELL · AZURE193**

```
cd /opt/services/satellite-status
docker compose ps
echo "SELECT @@global.time_zone;" | db     -- +00:00
```

## D. During the window: verify status.amsat.org

- [ ] Home page loads. The grid shows the last 6 days of real data from dallas190, and tooltips show exact times (`YYYY-MM-DD HH:MM UTC`).
- [ ] API version 1.1 and health:

  **SHELL · ANYWHERE**

  ```
  curl -si https://status.amsat.org/api/v1/health.php | grep -i -e x-amsat-api-version -e status
  # X-AMSAT-API-Version: 1.1   and   "status": "ok"
  ```

- [ ] Reports: `reported_time` has real minutes and there's no `period` field:

  **SHELL · ANYWHERE**

  ```
  curl -s 'https://status.amsat.org/api/v1/reports.php?hours=168&limit=3'
  ```

- [ ] Legacy endpoints and the versions index still respond:

  **SHELL · ANYWHERE**

  ```
  curl -s https://status.amsat.org/api/v1/satellites.php | head -c 200
  curl -s 'https://status.amsat.org/api/v1/sat_info.php?name=ISS_%5BFM%5D&hours=96' | head -c 200
  curl -s https://status.amsat.org/api/
  ```

- [ ] **Submit one real report** through the form on status.amsat.org (your own callsign). It appears in the grid tooltip with the exact minute, and in the DB with `submission_time` set:

  **SHELL · AZURE193**

  ```
  echo 'SELECT id, callsign, observation_time, submission_time
        FROM satellite ORDER BY id DESC LIMIT 1;' | db
  ```

  If it was only a test, delete that one row by its id.

- [ ] No errors in the logs:

  **SHELL · AZURE193**

  ```
  cd /opt/services/satellite-status
  docker compose logs --since 30m frontend api | grep -i -E 'error|fatal|warn' | tail
  ```

## E. During the window: switch traffic

- [ ] Enable the drafted **308** redirect rule in Cloudflare.

  Redirect enabled (UTC) _____

- [ ] Redirect checks (path, trailing slash, query string kept):

  **SHELL · ANYWHERE**

  ```
  # expect 308 and Location: https://status.amsat.org[/...]
  curl -sI https://www.amsat.org/status  | grep -i -e '^HTTP' -e '^location'
  curl -sI https://amsat.org/status/     | grep -i -e '^HTTP' -e '^location'
  curl -sI 'https://www.amsat.org/status/api/v1/reports.php?name=AO-91&hours=1' \
       | grep -i location
  ```

- [ ] **POST survives the redirect** (empty body, so nothing is stored; a 422 from the new API proves the POST arrived as a POST):

  **SHELL · ANYWHERE**

  ```
  curl -si -L -X POST https://www.amsat.org/status/api/v1/reports.php \
       -H 'Content-Type: application/json' -d '{}' | grep -i -e '^HTTP' -e missing_name
  # HTTP/2 308 ... then HTTP/2 422 with "missing_name"
  ```

- [ ] Unrelated amsat.org pages are **not** redirected (e.g. the home page, and a page with "status" in a longer path).

- [ ] Put the welcome banner on the new site (live on the next page load, no deploy needed):

  **SHELL · AZURE193**

  ```
  cp banner-welcome-azure193.md /opt/services/satellite-status/content/banner.md
  ```

- [ ] Optional: on dallas190, replace the maintenance banner with a one-line "We've moved to status.amsat.org". It's only seen by anyone bypassing the redirect.
- [ ] Cloudflare caching: decide whether to turn back on what you disabled while testing the banner.

Window end (UTC) _____

## F. After the window

### Same day

- [ ] Tell the API consumers it's live (new base URL, API version 1.1, CHANGELOG link). Update Reiner on #15.
- [ ] Confirm #24 closed automatically when #30 merged. Comment on #32 that 1.1 shipped.
- [ ] **#49** (health data freshness, closes #47) is a **draft marked DO NOT MERGE**: it's stacked on #30's branch. **Ask Claude to rebase it onto** `main` **and retarget it**, then mark it ready, wait for green CI, and merge. Then check `health.php` shows version 1.2 and `"status": "ok"` with a recent `data_freshness.latest_report_time`.

  **SHELL · ANYWHERE**

  ```
  curl -s https://status.amsat.org/api/v1/health.php
  ```

- [ ] Re-check logs and that new reports are arriving:

  **SHELL · AZURE193**

  ```
  echo 'SELECT COUNT(*) FROM satellite WHERE submission_time > NOW() - INTERVAL 6 HOUR;' | db
  ```

### First week

- [x] **#31**: grid vote counts only each callsign's latest report. **Done:** shipped in #30 and closed at merge.
- [ ] **#46**: make the status page queries use the new `observation_time` index (check with `EXPLAIN` first).
- [ ] Watch for API clients still calling the old URL (follow-the-redirect is fine; browser apps that rely on CORS preflight will fail and need the new base URL).

> **The 308 redirect is permanent.** It runs at Cloudflare's edge, costs nothing, and doesn't depend on dallas190, so links, bookmarks and hard-coded clients from the last two decades keep working indefinitely. Nothing on the old URL can show a banner after cutover, so the welcome banner on status.amsat.org is what tells humans to update bookmarks. Old-URL traffic is tracked so you can reach the clients still using it, not so the redirect can be retired.

### Ongoing (monthly)

- [ ] **Check old-URL traffic** in Cloudflare analytics: host `amsat.org` / `www.amsat.org`, path starts with `/status`, response status 308, broken down by user agent. Contact identifiable API clients still using it. `OPTIONS` requests on old paths are browser apps failing CORS preflight, so contact those first.

### End of burn-in (you choose the length)

- [ ] Remove the welcome banner from status.amsat.org (a few weeks after cutover).
- [ ] Retire `/status` on dallas190 (the edge redirect is unaffected). Archive the final dump and the `submit.php.cutover-bak` backup.
- [ ] Optional, if old-URL browser traffic stays high: a short "We've moved, update your bookmarks" interstitial for browser paths only. The API keeps the plain 308.
- [ ] Then: **#45** (API v1.3: `observation_time`/`submission_time` fields, closes #32) together with **#42 §4** (OpenAPI schemas).

### Reference

Repo: AMSAT-NA/satellite-status · PR #30 · Issues #24, #31, #32, #45, #46 · Host path: `/opt/services/satellite-status` on azure193 · DB commands run inside the db container using its own `MARIADB_*` env vars, so passwords aren't typed. Placeholders in `<angle brackets>` are yours to fill in.

## G. Rollback

> Before the redirect is enabled (sections B–D), rolling back costs nothing: dallas190 still has every report. After the redirect, reports submitted to azure193 exist only there, so export them before switching back.

- [ ] Disable the Cloudflare 308 rule (traffic returns to dallas190 immediately).
- [ ] Restore write privileges on dallas190 (`GRANT ALL PRIVILEGES ON amsat_status.* TO 'amsatstatus'@'localhost';`), restore the form handler (`sudo cp -p ~/submit.php.cutover-bak /var/www/www.amsat.org/status/submit.php`), and swap the banner back.
- [ ] If the redirect was ever enabled: export reports received on azure193 since cutover, for re-entry on dallas190 (old schema: `day` = date, `hour`, `period` 0–3 = minutes 0–15 / 16–30 / 31–45 / 46–59, as the old app mapped them):

  **SHELL · AZURE193**

  ```
  echo "SELECT name, callsign, report, grid_square,
    DATE(observation_time) day, HOUR(observation_time) hour,
    CASE WHEN MINUTE(observation_time) <= 15 THEN 0
         WHEN MINUTE(observation_time) <= 30 THEN 1
         WHEN MINUTE(observation_time) <= 45 THEN 2 ELSE 3 END period
  FROM satellite WHERE submission_time >= '<cutover UTC>';" | db
  ```

- [ ] azure193 can stay as it is for a second attempt. To undo just the database migration there, use the rollback script, but **only** before new reports exist (they have no day/hour/period).

  **SHELL · AZURE193**

  ```
  db < 2026_08_08_add_observation_time_submission_time_rollback.sql
  ```

- [ ] Revert #30 on `main` (GitHub "Revert" on the merged PR) only if azure193 must run the old code again.